

LEO LEONG

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Currently serving National Service, available for full-time work from 2nd August 2026

SUMMARY

Cybersecurity & Digital Forensics graduate from Temasek Polytechnic with a strong foundation in software development, networking, and cybersecurity. Built full-stack software projects independently, including a café e-commerce platform and an ML-based HDB resale price predictor. Experienced in automation tooling and proof-of-concept development from an internship at Trend Micro. Seeking a software engineering role from August 2026.

SKILLS

Programming: Python, Java, JavaScript, TypeScript, PHP, HTML/CSS, SQL

Frameworks & Tools: Next.js, React, Express.js, Sequelize ORM, MQTT.js, Nodemailer, SendGrid, JWT, bcrypt, Git

Cloud & Databases: Microsoft Azure, PostgreSQL, MySQL

Cybersecurity Tools: Burp Suite, Nmap, Metasploit, Kali Linux, Wireshark

Data & ML: scikit-learn, pandas, Streamlit, Power BI

Systems: Linux (Red Hat), Nginx, Android Studio

Certificates: AWS Academy Cloud Foundations

WORK EXPERIENCE

Technical Intern

April 2023 – September 2023

Trend Micro

- Developed POC documentation for Trend Micro security products by deploying controlled VM environments, injecting test malware, and validating detection, logging, and remediation behaviors. Produced reports adopted by Sales Engineers in enterprise client presentations
- Developed a Python automation tool that extracted client data (e.g., name, email, and product details) from Excel spreadsheets and populated a standardized email template with customer-specific information for outreach campaigns. Independently designed, tested, and debugged the solution, eliminating manual one-by-one email distribution for the Sales Engineering team.

Vehicle Commander, Terrex ICV

2025 – Present

Singapore Armed Forces

- Lead a crew during training and operational exercises, coordinating communication and decision-making to ensure safe and effective mission execution.
- Took ownership of mission planning, crew readiness, and real-time problem solving in dynamic environments, strengthening leadership, adaptability, and composure under pressure.

PROJECTS

EcoSaver — IoT Air Quality Monitoring Platform | *Node.js, Express, MySQL, Azure, MQTT, EJS, Nginx*

- Designed and built a full-stack air quality monitoring web platform integrating live IoT sensor data via MQTT, deployed on Microsoft Azure with a custom domain and SSL/TLS certificates
- Developed a consumer-facing website displaying real-time and historical air quality metrics (PSI, NO2, SO2, CO, O3, Temperature, Humidity) with daily and weekly data visualisation charts
- Built a separate admin management portal with dashboards for server uptime, database status, CPU monitoring, user management, sensor management, and API log tracking
- Designed and documented RESTful APIs secured with token-based authentication to support third-party integration with IoT devices and external applications.
- Configured a Linux production environment using Nginx reverse proxy, systemd services, and Let's Encrypt SSL certificates to ensure reliable and secure deployment.

Leo's Café — Full-Stack Café | *Node.js, Express, PostgreSQL, Sequelize ORM, JWT, bcrypt, SendGrid, QR Code, EJS*

- Designed and developed a full-stack café management platform supporting online ordering, table reservations, loyalty rewards, QR code redemption, and customer account management.
- Implemented a secure authentication and authorization system using JWT stored in httpOnly cookies, bcrypt password hashing, email verification, and rate limiting to protect user accounts and API endpoints.

- Designed a normalized PostgreSQL database with Sequelize ORM, implementing relational data models, input validation, and server-side sanitization to ensure data integrity and application security.
- Integrated automated email notifications using SendGrid for account verification and reservation confirmations, improving the overall customer experience.

HDB Resale Price Predictor | *Python, pandas, scikit-learn, Random Forest, Streamlit*

- Developed an end-to-end machine learning application that predicts HDB resale prices using historical transaction data from data.gov.sg.
- Performed data cleaning, feature engineering, and model training using Random Forest regression, achieving an R^2 score of 0.843 with a mean absolute error of S\$68,844.
- Deployed an interactive Streamlit web application allowing users to estimate future resale prices based on property characteristics.

Penetration Testing Engagement | *Nmap, Burp Suite, Metasploit, Kali Linux, Wireshark*

- Conducted a simulated professional penetration testing engagement applying the Cyber Kill Chain methodology to systematically identify vulnerabilities across a target network
- Used Nmap for network reconnaissance, Burp Suite for web application testing, and Metasploit for exploitation; produced a remediation report with CVSS-scored findings and recommended countermeasures

Company Network Infrastructure Design | *Cisco Packet Tracer, VLAN, IP Subnetting*

- Designed and configured WAN connectivity between Southeast Asia offices and overseas cloud facilities for a simulated MNC, with VLAN segmentation and IP subnetting across all sites

EDUCATION

Diploma in Cybersecurity & Digital Forensics | Temasek Polytechnic

April 2021 – May 2024

- Admitted into the Director's List for AY2021/2022 for excellent academic performance
- Recipient of Edusave Certificate of Academic Achievement, 2022
- Achieved Distinctions in Network Technology and Coding & Development Project

Cambridge GCE O-Level | Pasir Ris Secondary School

2016 – 2020

- Recipient of Edusave Scholarship (2018, 2020), Edusave Certificate of Academic Achievement (2019), and Li Kaiyan Progress Award (2020)